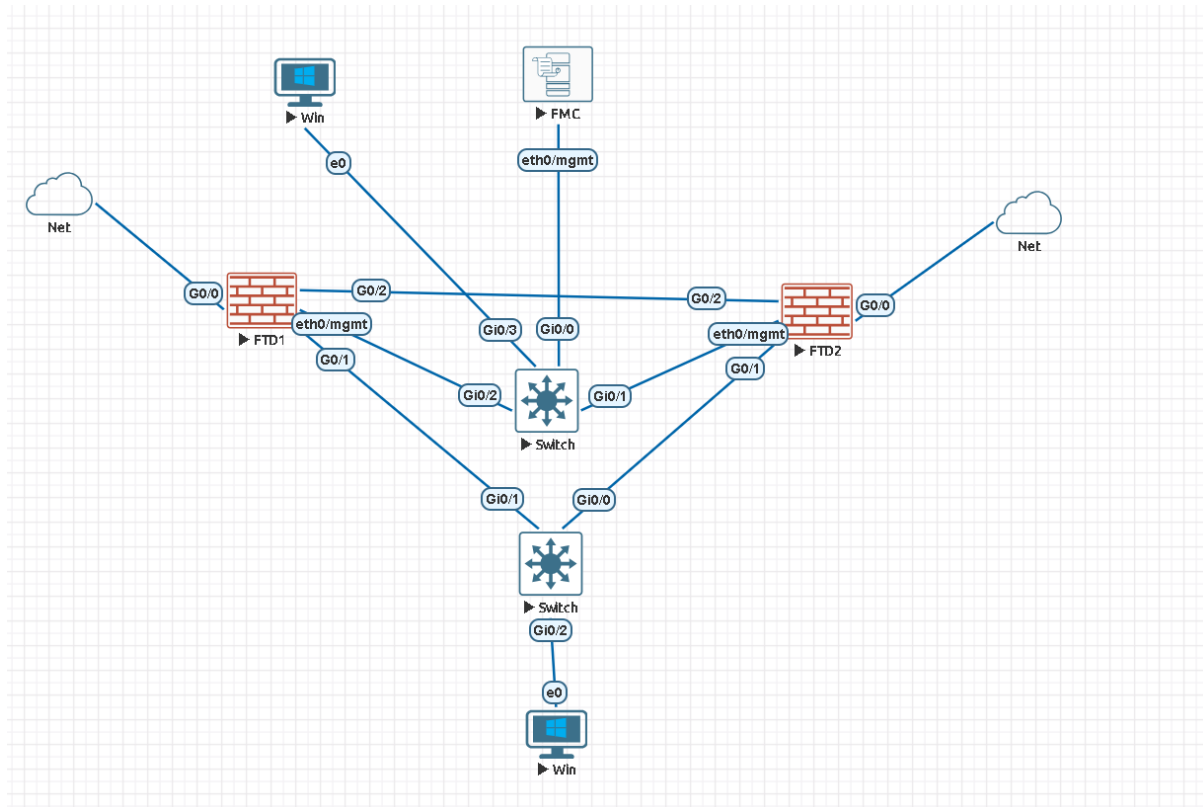


Abdelrhman nabil

OT cybersecurity engineer



☐	Name	Model	Ver...	Chassis	Licenses	Access Control Policy	Auto RollBack
☐	Ungrouped (2)						
☐	FTD1 Short 3 192.168.10.2 - Routed	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	⌂ ⚙
☐	FTD2 Short 3 192.168.10.3 - Routed	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	⌂ ⚙

FTD Cluster vs HA

- **HA (Active/Standby)**
 - Two FTDs
 - One Active, one Standby
 - Stateful failover only
 - Only one unit processes traffic at a time
- **FTD Cluster (Active/Active)**
 - Two or more FTDs
 - All units actively process traffic
 - Used for **scalability & throughput aggregation**
 - Connections are distributed across cluster members

Clustering is **different from HA**, but FMC manages both.

Step 1 — Prepare FTDs for Clustering

1. Make sure **inside and outside interfaces are identical** across both devices:
 - Zones: Inside/Outside
 - IPs/subnets
 - Default routes
 - NAT rules
2. Prepare a **cluster control link** (can be g0/2):
 - Assign IPs for each FTD:
 - FTD1 → 192.168.250.1/30
 - FTD2 → 192.168.250.2/30
 - This link is used for cluster communication, similar to HA failover, but for cluster control

Step 2 — Create Cluster in FMC

1. Go to:
Devices → Device Management → Add → Cluster

The screenshot shows the Firewall Management Center (FMC) interface. The top navigation bar includes 'Home', 'View By: Group', a search bar, and a 'Deploy' button with a notification icon showing 2 alerts. The left sidebar contains navigation options: Overview, Analysis, Policies, Devices (selected), and Objects. The main content area displays a table of devices. A dropdown menu is open over the 'Add' button, showing options: Device, Device (Wizard), Cluster (highlighted), Chassis, High Availability, and Group. The table below shows two devices, FTD1 and FTD2, both with status 'Normal' and 'Routed'.

Name	Model	Ver...	Chassis	Licenses	Access Control Policy	Av
FTD1 Short 3 192.168.10.2 - Routed	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	⏪ ⏩ ⚙ ⋮
FTD2 Short 3 192.168.10.3 - Routed	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	⏪ ⏩ ⚙ ⋮

2. Enter:

Add Cluster Wizard

1 Configuration — 2 Summary

 Create a cluster for supported models. Note: For the Firepower 4100/9300 and threat defense virtual (AWS/GCP/Azure), use the Add Device option. Make sure connected switches match the MTUs for data interfaces and the cluster control link interface.

Cluster Name *

Cluster-Test

Cluster Key

.....

.....

Control Node

You can form the cluster with just the control node to reduce formation time.

Node *

FTD2

VXLAN Network Identifier (VNI) Network

10.10.10.0

/ 27 (30 addresses)

Virtual Tunnel Endpoint (VTEP) Network

10.10.20.0

/ 27 (30 addresses)

Cluster Control Link *

GigabitEthernet0/2

VTEP IPv4 Address *

10.10.20.1

Priority *

1

Data Nodes (Optional)

Data node hardware needs to match the control node hardware.

[Add a data node](#)

Activate Windows
Go to Settings to activate Windows.
[Cancel](#) [Continue](#)

The Following is Explanation of each field:

1. Cluster Name

This is simply the **logical name** of the cluster object inside FMC.

- Used only in FMC for identification
- Does NOT affect the FTDs or the network
- You can name it anything

Example:

Cluster-Test or FTD-Cluster-LAB

2. Cluster Key

This is the **encryption key** used to secure and authenticate cluster communication between members.

Why it exists?

- All cluster members must use the same shared secret
- Prevents unauthorized nodes from joining the cluster
- Encrypts control-plane sync over the CCL (Cluster Control Link)

How to fill it:

- Put any password-like string
- Same Key must be entered twice (confirm)

Example:

Cisco123!

3. Control Node → Node

This dropdown chooses **which FTD becomes the cluster CONTROL NODE** during creation.

What is a Control Node?

- The **first member** you add to the cluster
- Manages initial cluster formation
- Holds the initial VXLAN VNI address assignments
- Other members join it

You should choose the FTD that:

- Is online
 - Is reachable by FMC
-

4. VXLAN Network Identifier (VNI) Network

What is this?

This is the **internal virtual network** used inside the cluster to synchronize data using VXLAN.

The cluster uses **VXLAN tunnels** between members to:

- Sync NAT tables
- Sync connection states
- Sync routing and cluster buckets
- Manage traffic distribution

VNI Network = a **virtual network** reserved for internal cluster communication

This network does **not exist physically** on your switches or LAN.

It is fully internal.

Format:

<VNI network IP> / <Prefix>

Requirements:

- Must be unique
- Must NOT overlap with any existing real network
- Must NOT be routable in your network
- Best practice: use reserved/not-used ranges like 10.10.x.x

Example:

10.10.10.0 / 27

5. Virtual Tunnel Endpoint (VTEP) Network

This is also a **virtual network** used for VXLAN encapsulation endpoints.

VXLAN relies on VTEPs (Virtual Tunnel Endpoints).

Each node gets a VTEP address.

Requirements:

- Must also be unique
 - Must NOT overlap with any real network
 - Best practice: use another unique network
-

6. VTEP IPv4 Address

This is the **VTEP address for the Control Node** inside the VTEP network.

How to select it?

It must be the first usable IP inside the VTEP Network.

Example:

If VTEP Network = 10.10.20.0/27

- Network: 10.10.20.0
- First usable: 10.10.20.1 → this goes to Control Node (FTD1)

So you fill:

10.10.20.1

The second member will get:
10.10.20.2 automatically when you add them.

7. Cluster Control Link

This is the **physical interface used for cluster synchronization**.

This is where:

- Connection tables sync
- NAT tables sync
- VXLAN tunnels run
- Cluster heartbeats occur

Important:

This must be the **dedicated direct cable** between your FTDs.

In your lab:

✓ g0/2 is the correct CCL interface

So choose:

GigabitEthernet0/2

8. Priority

This defines which node has preference to become **Master / Control Member** in the cluster.

Rules:

- Lower number = higher priority
- Node with priority 1 becomes preferred master
- If it fails, next lowest wins

Recommended:

- Keep control node at 1
- Additional nodes: 5, 10, etc.

For you:

1 (default is best)

WHY FMC NEEDS ALL THESE FIELDS (full explanation)

To create a cluster, FMC builds:

- VXLAN mapping
- Internal tunnel IPs
- Virtual MAC calculation
- Bucket distribution table
- Control-plane sync mappings

These require:

- VNI → VXLAN overlay network
- VTEP → Tunnel endpoints
- CCL → Physical transport
- Priority → Master selection
- Cluster Key → Encryption & authentication

All these are mandatory for the cluster to operate as an **Active/Active distributed firewall**.

Add Cluster Wizard

1 Configuration — 2 Summary

⚠️ Clustering requires jumbo frames for the cluster control link. If you did not enable jumbo frames at deployment or have not previously enabled jumbo frames by setting the MTU of an interface above 1500, you need to manually reboot each node after the cluster is formed and healthy. Use the 'show jumbo-frame reservation' command on the device to check jumbo frame status.

 The cluster control link uses a higher MTU than data interfaces. The current max data MTU is 1500; the cluster control link MTU will be set to 1654. Make sure you set the switch MTU to match, otherwise cluster formation will fail. [More info](#)

Cluster Name: **Cluster-Test**

Cluster Key: (Specified)

Control Node

Name	Priority	VNI Network	VTEP IPv4 Address	Cluster Control Link	VTEP Network
FTD2	1	10.10.10.0/27	10.10.20.1	GigabitEthernet0/2	10.10.20.0/27

3. FMC will:

- Create logical cluster object
- Push cluster configuration to both members

Cluster-Test (2)									
Cluster (Individual Interface Mode)									
FTD1	Short 3	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	N/A		
192.168.10.2 - Routed									
FTD2	Short 3	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	N/A		
192.168.10.3 - Routed									

Step 4 — Deploy Configuration

- Click **Deploy** → **Deploy Now**
- Both FTDs will reboot as part of cluster formation
- Cluster state will appear as:
 - **Member 1** → **Active**
 - **Member 2** → **Active**
 - Traffic load balanced (per configured cluster algorithm)

```
Cluster Cluster-Test: On
Interface mode: individual
Cluster Member Limit : 16
This is "FTD1" in state DATA_NODE
  ID       : 1
  Version  : 9.19(1)200
  Serial No.: 9AAFMG286F5
  CCL IP   : 10.10.10.2
  CCL MAC  : 5000.0002.0004
  Module   : NGFWv
  Resource : 4 cores / 8192 MB RAM
  Last join : 19:19:09 UTC Nov 25 2025
  Last leave: N/A
Other members in the cluster:
Unit "FTD2" in state CONTROL_NODE
  ID       : 0
  Version  : 9.19(1)200
  Serial No.: 9AUQHL9LQFG
  CCL IP   : 10.10.10.1
  CCL MAC  : 5000.0003.0004
  Module   : NGFWv
  Resource : 4 cores / 8192 MB RAM
  Last join : 19:15:17 UTC Nov 25 2025
  Last leave: N/A
```

Step 5 — Configure Policies for Cluster

1. **Access Control Policies (ACP)** → assign to cluster
2. **NAT Rules** → assign to cluster
3. **Routing** → ensure cluster interface IPs are reachable

Once deployed, the **cluster acts as a single logical firewall**, all policies applied centrally.

Step 6 — Verification

- **FMC → Device Management → Cluster → Status**
 - Both members online
 - Cluster Control Link green
 - Sync Status: Completed
-

Why Clustering?

1 Performance and Scalability

- Cluster distributes flows across multiple FTD nodes.
- Increases overall throughput, concurrent sessions, and connection-per-second (CPS) capacity.
- Provides horizontal scaling – add more appliances as needed.

2 High Availability

- If a node fails, remaining nodes continue handling existing and new traffic.
- Better than classic HA which only uses one appliance at a time.

3 Single Logical Device

- FMC manages the cluster as one firewall.
 - NAT, Access-Control, Routing, VPN, and Policies are applied cluster-wide.
-